SCENARIOS

USE CASE SCENARIOS – HIDROSANITARE AGOLLI SYSTEM

Project: Order, Stock, Customer and Supplier Management

Company: Hidrosanitare Agolli

SCENARIO 1: Register a new order from an existing customer

Name: Register order

Primary actor: Salesperson

Preconditions: Customer exists in the system; products are registered

Postconditions: Order is created with status "Pending"; stock is temporarily reserved

Normal flow:

1. Salesperson opens "Orders" module and selects "Create new order"

2. System asks for customer ID or name

3. Salesperson selects a customer from the displayed list

4. System shows an empty order and asks for delivery details (address, desired date)

5. Salesperson enters products using product name or code and quantity

6. System checks stock: if sufficient, adds product to order and calculates subtotal

7. Salesperson repeats steps 5-6 for each product

8. Salesperson confirms the order

9. System calculates total, taxes (if any), creates order with status "Confirmed" and generates order number

10. System prints or displays order confirmation

Alternative flow (insufficient stock):

At step 6, system notifies salesperson: "Insufficient stock – only X units available". Salesperson can reduce quantity or remove the product.

SCENARIO 2: Create a new customer during an order

Name: Register new customer

Primary actor: Salesperson

Preconditions: Customer does not exist in the system

Postconditions: New customer is registered; his order is linked to him

Normal flow:

1. During order creation (step 2 of Scenario 1), salesperson selects "New customer"

2. System displays customer data form: name, address, phone, email, type (Business/Individual)

3. Salesperson fills in the data

4. System checks if email or phone already exists – if not, creates customer and assigns unique ID

5. System continues with order creation

SCENARIO 3: Process order and generate invoice

Name: Process order and generate invoice

Primary actor: Manager or automated system

Preconditions: Order has status "Confirmed"; stock is sufficient (or reserved)

Postconditions: Order status becomes "Completed"; stock is updated; invoice is created and sent by email

Normal flow:

1. Manager (or scheduled process) opens the list of confirmed orders

2. Selects an order and clicks "Process"

3. System verifies stock for each product in the order

4. System decreases stock for each product

5. System generates an invoice containing: invoice number, date, customer details, list of products with unit prices and totals, total amount, VAT

6. System changes order status to "Completed"

7. System sends invoice by email to the customer (if email exists) and stores it in data store D5

8. System notifies manager of completion

Alternative flow (insufficient stock):

At step 3, system detects insufficient stock for a product. System sets order status to "Awaiting supply" and automatically creates a purchase order for that product (see Scenario 5).

SCENARIO 4: Check stock and alert for low stock

Name: Check stock and generate alert

Primary actor: Scheduled system / Warehouse worker

Preconditions: Products have defined minimum stock levels

Postconditions: A list of low-stock products is created; a purchase order may be generated

Normal flow:

1. Every morning at 08:00, the system runs a process that checks stock of all products

2. For each product, system compares quantity in stock with minimum level (e.g., 10 units)

3. If quantity in stock is less than or equal to minimum level, the product is added to the "low stock" list

4. System generates a report and emails it to the warehouse worker and manager

5. Warehouse worker opens the report and decides which products need to be ordered

Alternative flow (automatic order):

In system configuration, "Automatic order" is enabled. Once a product is identified as low stock, the system automatically creates a purchase order to the primary supplier (see Scenario 5).

SCENARIO 5: Create and process a purchase order (to supplier)

Name: Order products from supplier

Primary actor: Warehouse worker

Preconditions: Product exists; its supplier is registered

Postconditions: A purchase order is created; when goods arrive, stock is increased

Normal flow:

1. Warehouse worker opens "Suppliers" module and selects "Create purchase order"

2. System shows the list of low-stock products (from Scenario 4)

3. Warehouse worker selects products and desired quantities

4. System automatically selects the primary supplier for each product (if multiple suppliers, worker chooses)

5. Warehouse worker confirms the purchase order

6. System generates a purchase order document and saves it

7. Warehouse worker sends the order by email or prints it to send to the supplier

8. When products arrive, warehouse worker opens the order and clicks "Receive shipment"

9. System adds quantities to stock and records the receipt date

SCENARIO 6: Product return from customer and refund

Name: Process return

Primary actor: Salesperson or Manager

Preconditions: A completed order exists for that customer; products are in returnable condition

Postconditions: Stock is increased (if product is resalable); a credit note is created

Normal flow:

1. Customer presents the product and the original invoice

2. Salesperson opens "Orders" module and finds the original order

3. Salesperson selects "Return product"

4. System displays the list of products from that order

5. Salesperson selects the product(s) to return and the quantity

6. System calculates the refund amount (based on selling price at the time of that order)

7. Salesperson chooses refund method (cash, store credit, bank transfer)

8. System generates a "Credit Note" and gives it to the customer

9. System increases stock for returned products (if resalable) or marks them as "damaged"

10. System records the return and links it to the original order

SCENARIO 7: Generate weekly sales reports

Name: Weekly sales report

Primary actor: Manager

Preconditions: At least some completed orders exist

Postconditions: Report is displayed on screen and can be exported to PDF or Excel

Normal flow:

1. Manager opens "Reports" module, then "Sales", then "Weekly"

2. System asks for date range (last week, or specific dates)

3. Manager selects the interval and clicks "Generate"

4. System collects all completed orders within that interval

5. System calculates: total sales, number of orders, best-selling products, sales by product category, sales by salesperson (employee who registered orders)

6. System displays the report in tabular and graphical form (bar chart)

7. Manager can export the report as PDF or Excel

SCENARIO 8: Register a new employee and assign a supervisor

Name: Register new employee

Primary actor: Administrator

Preconditions: Administrator is logged in

Postconditions: New employee has an account to log into the system

Normal flow:

1. Administrator opens "Administration" module, then "Employees", then "Add employee"

2. System displays form: name, position, hire date, phone, email, role (Salesperson/Warehouse/Manager/Admin)

3. Administrator enters data and selects a supervisor (another employee) from the list

4. System verifies that the email has not been used before

5. System creates an account with an automatically generated password (sent by email to the employee)

6. Employee logs in and changes the initial password

SCENARIO 9: Cancel an order before processing

Name: Cancel order

Primary actor: Salesperson (or Customer through salesperson)

Preconditions: Order has status "Pending" or "Confirmed", but not yet processed

Postconditions: Order is cancelled; reserved stock is released

Normal flow:

1. Salesperson finds the order in the list of active orders

2. Salesperson selects the order and clicks "Cancel"

3. System asks for confirmation: "Are you sure you want to cancel this order?"

4. Salesperson confirms

5. System changes order status to "Cancelled"

6. System releases reserved stock (if any reservation existed)

7. System records the reason for cancellation (optional)

SCENARIO 10: Login and role-based access

Name: Login and role-based access

Primary actor: Any user (Salesperson, Warehouse, Manager, Admin)

Preconditions: User has an active account

Postconditions: User sees only the modules allowed for their role

Normal flow:

1. User opens login page and enters email and password

2. System verifies credentials

3. System loads the user's role (salesperson, warehouse, manager, admin)

4. System displays the main dashboard with allowed modules:

- **Salesperson: Customers, Orders, Invoices, Simple reports**
- **Warehouse: Products, Stock, Suppliers, Purchase orders**
- **Manager: All modules except user administration**
- **Admin: All modules + user and role management**

5. User works within their permissions

6. If user tries to access a module without permission, system shows error "You are not authorized"

SCENARIO 11: Change the selling price of a product

Name: Update product price

Primary actor: Manager

Preconditions: Product exists

Postconditions: New price is used for future orders

Normal flow:

1. Manager opens "Products" module and then "Product list"

2. Finds the product and clicks "Edit"

3. System displays the edit form

4. Manager changes the "selling price" field and gives a reason for the change (optional)

5. System asks for confirmation

6. Manager confirms

7. System logs the change and updates the product

8. Manager may choose to apply the new price also to pending orders (in special cases)

SCENARIO 12: Record receipt of a shipment (stock update)

Name: Receive shipment

Primary actor: Warehouse worker

Preconditions: A purchase order has been sent; goods have physically arrived

Postconditions: Stock is increased; purchase order is marked as "Completed"

Normal flow:

1. Warehouse worker opens "Suppliers" module, then "Purchase orders", then "In transit"

2. Finds the order that has arrived and clicks "Receive shipment"

3. System displays the list of products and ordered quantities

4. Warehouse worker enters the actual quantity received (may be smaller if some items are missing)

5. For each product, warehouse worker can choose to put into stock or place in "quarantine" if quality is doubtful

6. System adds the confirmed quantity to quantity in stock

7. System changes the status of the purchase order to "Completed"

8. System records the receipt date and the person who processed it

9. If the purchase order was automatically created due to low stock, the system removes the low stock alert for those products